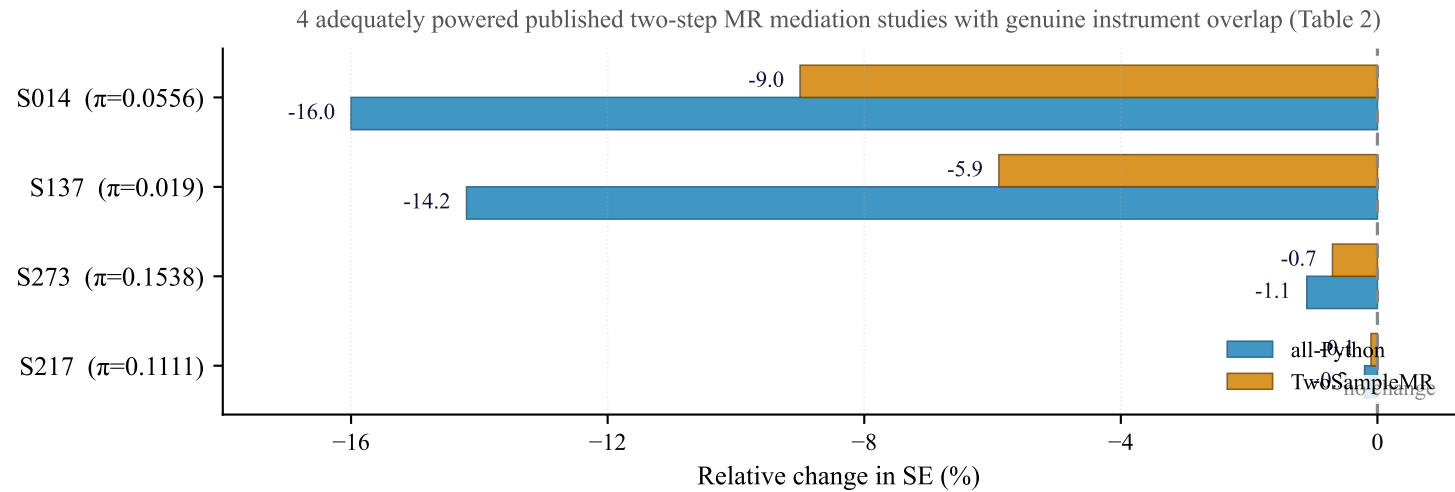


**Supplementary Figure S4. Change in indirect-effect SE after the covariance correction: all-Python batch vs TwoSampleMR cross-check**



Both pipelines narrow; TwoSampleMR returns a systematically smaller correction (−0.1% to −9.0%) than the all-Python batch (−0.2% to −16.0%, median −1.6%) because the wrappers re-harmonise effects differently (e.g., S014  $\hat{\alpha}$  SE 0.00096 vs 0.00345).